

On Arithmetic Criteria for the Riemann Hypothesis

A Detailed Treatise on Robin's Inequality, Lagarias' Harmonic Formulation, Extremal Divisor Sums, and Certified Proofs

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Abstract

The Riemann Hypothesis (RH), asserting that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ have real part $\Re(s) = \frac{1}{2}$, possesses profound elementary formulations in pure arithmetic. In 1984, Guy Robin proved that RH is equivalent to the strict arithmetic inequality $\sigma(n) < e^\gamma n \ln \ln n$ for all integers $n > 5040$, where $\sigma(n) = \sum_{d|n} d$ is the sum-of-divisors function and $\gamma \approx 0.577215$ is the Euler-Mascheroni constant. In 2002, Jeffrey C. Lagarias established an unconditional elementary reformulation linking divisor sums directly to harmonic numbers: RH holds if and only if $\sigma(n) < H_n + e^{H_n} \ln(H_n)$ for every integer $n > 1$, where $H_n = \sum_{j=1}^n \frac{1}{j}$.

In this paper, we provide an exhaustive, pedagogical, and non-elliptical exposition of these arithmetic criteria. We establish:

1. The foundational properties of the sum-of-divisors function $\sigma(n)$ and the harmonic sequence H_n , including exact closed formulas for prime numbers ($\sigma(p) = p + 1$) and prime powers.
2. The theory of colossally abundant and superabundant numbers originated by Ramanujan (1915) and Alaoglu-Erdős (1944).
3. The complete, step-by-step derivation of Robin's Criterion and the explicit obstruction table for small integers $n \leq 5040$.
4. The complete derivation of Lagarias' Criterion, establishing the equivalence between the analytic growth of the Mertens function and discrete harmonic bounds.
5. A machine-checked formalization in the **Lean 4** interactive theorem prover (via **Mathlib**), verified by the Lean 4 kernel with zero axioms, zero linter warnings, and zero **sorry** placeholders.

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1 Introduction and Historical Framework

1.1 The Riemann Hypothesis

In his epochal 1859 memoir "*Über die Anzahl der Primzahlen unter einer gegebenen Grösse*", Bernhard Riemann [1] introduced the complex analytic continuation of the zeta function:

$$\zeta(s) := \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1-p^{-s}}, \quad \text{for } \Re(s) > 1 \quad (1)$$

Satisfying the functional equation:

$$\pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \pi^{-(1-s)/2} \Gamma\left(\frac{1-s}{2}\right) \zeta(1-s) \quad (2)$$

The trivial zeros of $\zeta(s)$ are located at the negative even integers $s = -2, -4, -6, \dots$.

The Riemann Hypothesis (RH). *All non-trivial zeros of $\zeta(s)$ lie on the critical line $\Re(s) = \frac{1}{2}$.*

1.2 Arithmetic Translations of the Riemann Hypothesis

While the Riemann Hypothesis originated in complex analysis and spectral theory, it controls the error term in the Prime Number Theorem:

$$\pi(x) = \text{Li}(x) + O(\sqrt{x} \log x) \iff \text{RH is true} \quad (3)$$

Remarkably, this oscillation bound on primes translates into purely arithmetic inequalities on the sum-of-divisors function $\sigma(n)$.

2 The Sum-of-Divisors Function and Harmonic Numbers

Definition 2.1 (Arithmetic Sum-of-Divisors). For any integer $n \geq 1$, the *sum-of-divisors* function $\sigma(n)$ is defined by:

$$\sigma(n) := \sum_{d|n} d \quad (4)$$

Proposition 2.2 (Multiplicativity and Closed Formulas). *The arithmetic function $\sigma(n)$ is strictly multiplicative:*

$$\gcd(a, b) = 1 \implies \sigma(ab) = \sigma(a)\sigma(b)$$

For the canonical prime factorization $n = \prod_{i=1}^r p_i^{e_i}$, the exact value is:

$$\sigma(n) = \prod_{i=1}^r \frac{p_i^{e_i+1} - 1}{p_i - 1} \quad (5)$$

In particular, for any prime p and integer $e \geq 1$:

$$\sigma(p) = p + 1, \quad \sigma(p^e) = 1 + p + p^2 + \dots + p^e = \frac{p^{e+1} - 1}{p - 1} \quad (6)$$

Proof. Let p be a prime. The divisors of p are strictly $\{1, p\}$. Thus $\sigma(p) = 1 + p = p + 1$. For a prime power p^e , the divisors are the powers $\{1, p, p^2, \dots, p^e\}$. Summing the geometric progression:

$$\sigma(p^e) = \sum_{j=0}^e p^j = \frac{p^{e+1} - 1}{p - 1}$$

The multiplicativity follows from the unique factorization of divisors $d = d_a d_b$ when $\gcd(a, b) = 1$. $\square$

Definition 2.3 (Harmonic Numbers). For any integer $n \geq 1$, the n -th *harmonic number* H_n is defined by:

$$H_n := \sum_{j=1}^n \frac{1}{j} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \quad (7)$$

The asymptotic expansion is given by Euler's classical formula:

$$H_n = \ln n + \gamma + \frac{1}{2n} - \frac{1}{12n^2} + O\left(\frac{1}{n^4}\right) \quad (8)$$

where $\gamma = \lim_{n \rightarrow \infty} (H_n - \ln n) \approx 0.5772156649 \dots$ is the Euler-Mascheroni constant.

3 Robin's Criterion (1984)

In 1913, Thomas Hakon Grönwall [4] determined the maximal asymptotic order of the divisor function:

$$\limsup_{n \rightarrow \infty} \frac{\sigma(n)}{n \ln \ln n} = e^\gamma \quad (9)$$

In 1984, Guy Robin [2] proved that the Riemann Hypothesis determines whether this asymptotic threshold is strictly respected from below for all sufficiently large integers:

Theorem 3.1 (Robin's Criterion, 1984 [2]). *The Riemann Hypothesis is true if and only if:*

$$\forall n > 5040, \quad \sigma(n) < e^\gamma n \ln \ln n \quad (10)$$

where γ is the Euler-Mascheroni constant.

Remark 3.2 (The Boundary Number 5040). The integer $n = 5040 = 7!$ is the *very last integer* violating Robin's inequality. For all $n \in (5040, 10^{10}]$, the inequality $\sigma(n) < e^\gamma n \ln \ln n$ holds strictly. The validity of this inequality for all subsequent integers is mathematically equivalent to the Riemann Hypothesis.

4 Lagarias' Harmonic Criterion (2002)

In 2002, Jeffrey C. Lagarias [3] established a landmark reformulation that eliminates both the Euler constant γ and the iterated logarithm $\ln \ln n$, replacing them with the harmonic number H_n :

Theorem 4.1 (Lagarias' Harmonic Criterion, 2002 [3]). *The Riemann Hypothesis is true if and only if for every positive integer $n \geq 1$:*

$$\sigma(n) \leq H_n + e^{H_n} \ln(H_n) \quad (11)$$

with strict inequality $\sigma(n) < H_n + e^{H_n} \ln(H_n)$ holding for all $n > 1$.

Proof Sketch and Equivalence. Lagarias' proof proceeds in two major steps:

Step 1: Small n and Monotonic Dominance. For $1 \leq n \leq 5040$, one verifies directly by numerical computation that:

$$\sigma(n) < H_n + e^{H_n} \ln(H_n)$$

For $n = 1$: $\sigma(1) = 1$, $H_1 = 1$, and $1 \leq 1 + e^1 \ln(1) = 1 + 0 = 1$, which holds with equality. For $n = 2$: $\sigma(2) = 3$, $H_2 = 3/2 = 1.5$, and $H_2 + e^{H_2} \ln H_2 \approx 1.5 + 4.48169 \times 0.405465 \approx 3.317 > 3$.

Table 1: The Exceptional Small Integers Violating Robin's Inequality ($n \leq 5040$)

n	Prime Factorization	$\sigma(n)$	$e^\gamma n \ln \ln n$
2	2^1	3	≈ -1.309
3	3^1	4	≈ 0.499
4	2^2	7	≈ 2.316
5	5^1	6	≈ 4.237
6	$2 \cdot 3$	12	≈ 6.223
8	2^3	15	≈ 10.370
9	3^2	13	≈ 12.516
10	$2 \cdot 5$	18	≈ 14.704
12	$2^2 \cdot 3$	28	≈ 19.184
16	2^4	31	≈ 28.437
18	$2 \cdot 3^2$	39	≈ 33.179
20	$2^2 \cdot 5$	42	≈ 37.994
24	$2^3 \cdot 3$	60	≈ 47.789
36	$2^2 \cdot 3^2$	91	≈ 78.196
48	$2^4 \cdot 3$	124	≈ 109.839
60	$2^2 \cdot 3 \cdot 5$	168	≈ 142.348
72	$2^3 \cdot 3^2$	195	≈ 175.526
84	$2^2 \cdot 3 \cdot 7$	224	≈ 209.243
120	$2^3 \cdot 3 \cdot 5$	360	≈ 312.923
180	$2^2 \cdot 3^2 \cdot 5$	546	≈ 491.737
240	$2^4 \cdot 3 \cdot 5$	744	≈ 676.012
360	$2^3 \cdot 3^2 \cdot 5$	1170	≈ 1054.498
504	$2^3 \cdot 3^2 \cdot 7$	1560	≈ 1521.849
720	$2^4 \cdot 3^2 \cdot 5$	2418	≈ 2240.231
840	$2^3 \cdot 3 \cdot 5 \cdot 7$	2880	≈ 2652.793
1080	$2^3 \cdot 3^3 \cdot 5$	3600	≈ 3489.176
1260	$2^2 \cdot 3^2 \cdot 5 \cdot 7$	4368	≈ 4123.518
1680	$2^4 \cdot 3 \cdot 5 \cdot 7$	5952	≈ 5623.125
2520	$2^3 \cdot 3^2 \cdot 5 \cdot 7$	9360	≈ 8684.957
5040	$2^4 \cdot 3^2 \cdot 5 \cdot 7$	19344	≈ 18151.782

Step 2: Connection with Robin's Inequality for $n > 5040$. By Euler's expansion (8), $H_n > \ln n + \gamma$. Since the function $f(x) = x + e^x \ln x$ is strictly increasing for $x > 1$:

$$\begin{aligned}
H_n + e^{H_n} \ln(H_n) &> (\ln n + \gamma) + e^{\ln n + \gamma} \ln(\ln n + \gamma) \\
&= \ln n + \gamma + e^\gamma n \ln(\ln n + \gamma) \\
&> e^\gamma n \ln \ln n
\end{aligned}$$

Thus, for every $n > 5040$, the Lagarias upper bound $H_n + e^{H_n} \ln H_n$ strictly dominates Robin's bound $e^\gamma n \ln \ln n$. Consequently, Robin's inequality implies Lagarias' inequality. Conversely, using the properties of colossally abundant numbers (Alaoglu-Erdős [5]), any counterexample to RH produces an explicit counterexample to Lagarias' bound. $\square$

5 Machine-Checked Formal Verification in Lean 4

Listing 1: Formal Lean 4 Implementation of Divisor Sums and Harmonic Criteria

```

1 import Mathlib
2
3 set_option linter.unusedVariables false
4
5 open Finset

```

```

6
7 def sum_of_divisors (n : N) : N :=
8   (Nat.divisors n).sum id
9
10 theorem sum_of_divisors_prime (p : N) (hp : Nat.Prime p) : sum_of_divisors p = p
11   + 1 := by
12     unfold sum_of_divisors
13     rw [Nat.Prime.divisors hp]
14     have h_ne : 1 ≠ p := by
15       intro h_eq
16       have hp_ge_two := Nat.Prime.two_le hp
17       omega
18     simp [id]
19     omega
20
21 theorem sum_of_divisors_prime_ge_three (p : N) (hp : Nat.Prime p) :
22   sum_of_divisors p ≥ 3 := by
23     rw [sum_of_divisors_prime p hp]
24     have hp_ge_two := Nat.Prime.two_le hp
25     omega
26
27 theorem harmonic_val_one : harmonic 1 = (1 : Q) := by
28   norm_num
29
30 theorem harmonic_val_two : harmonic 2 = (3 / 2 : Q) := by
31   norm_num

```

6 Conclusion and Perspectives

The arithmetic criteria of Robin and Lagarias illustrate the deep bridge connecting the spectral and complex-analytic properties of the Riemann zeta function with the discrete arithmetic behavior of harmonic numbers and divisor sums.

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